



Santhosh Kumar Ravi Kumar

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EDUCATION

Technical University of Munich

MSc in Biomedical Engineering and Medical Physics | June 2024 | Munich, Germany | German Grade: 2

National Institute of Technology, Raipur, India

B. Tech in Biomedical Engineering | June 2021 | Raipur, India | German Equivalent Grade: 2.1

National Institute of Technology, Warangal, India

PG Diploma in Artificial Intelligence and Machine Learning | March 2021 | Warangal, India | German Equivalent Grade: 1.3

Experience and Projects

(Mar 2025 – Present)

Machine Learning Scientific Assistant | Ferdinand Braun Institute, Berlin:

- Designing and deploying an interactive web application for facet defect inspection using Faster R-CNN with ResNet backbone and Region Proposal Networks, enabling precise multi-class defect detection via bounding box regression.
- Engineered a Flask-based backend integrated with OpenSeadragon and DeepZoom image pyramids, allowing fluid exploration of ultra high-resolution microscopy images with dynamic filtering and zooming.
- Developing scalable image preprocessing pipelines involving automated chunking, tiling, augmentation, and annotation to enhance model robustness and reduce memory bottlenecks during deep learning training.
- Collaborated with interdisciplinary teams to align ML tool development with scientific goals, delivering a production-ready prototype for internal deployment and active use in semiconductor materials analysis.

(Jun 2024 – Sep 2024)

Data Science Intern | Boehringer Ingelheim:

- Engineered and integrated advanced algorithms, Python packages, and Large Language Models (LLMs) to extract key information, including tables, from clinical trial reports, automating complex multi-page data extraction and reducing manual effort from 6 hours to 30 minutes.
- Developed an online application for PDF uploads, result visualization, and data export in multiple formats, enhancing data extraction efficiency and accuracy, thereby optimizing research workflows.
- Utilized chat-based LLMs for interactive and accurate information extraction, streamlining processes for scientists and facilitating more effective data analysis.

(Oct 2022 – May 2024)

Software Developer | Helmholtz Munich:

- Designed and implemented Calimera, a visualization software for the Institute of Diabetes and Obesity's Computational Discovery research group, facilitating the statistical and machine learning analysis of indirect calorimetry data from mouse studies.
- Built the platform using Python, Qt-Designer, and Shiny for Python, enabling researchers to efficiently interpret complex data, thereby accelerating the pace of diabetes and obesity research.
- Enhanced the research process by streamlining data analysis workflows, improving the accuracy and speed of deriving insights and conclusions.

PathoAICockpit | TUM DATA INNOVATION LAB:

Developed a web-based platform for real-time processing and quality control of digital pathology images using CNN models with TensorFlow and scikit-learn. Implemented algorithms for artifact detection, classification, and prediction, significantly improving diagnostic accuracy and processing efficiency in whole slide image analysis.

Masters Thesis: Autoencoder for Disease Prediction in Mice Models (TUM & Helmholtz):

Developed an RNN/LSTM autoencoder to analyze temporal indirect calorimetry data from mice, enabling disease prediction in knockout models based on reconstruction loss. Applied representation learning to mitigate batch effects and improve classification accuracy, showcasing the potential of deep learning for early disease detection in biomedical research.

Artificial Intelligence and Machine Learning (PG Diploma):

Gained practical experience in Python for data analysis and visualization using NumPy, Pandas, and Matplotlib. Designed workflows for supervised, unsupervised, and reinforcement learning, and implemented deep learning models including CNNs, RNNs, and GANs for applications in text classification, forecasting, and image recognition.

Additional Courses

- Electrical Computer Aided Design (ECAD), MATLAB, C++, Embedded development, SDLC, Requirements Engineering (IREB), Power BI, MySQL

SKILLS

Programming & Development:	Python, C++, MATLAB, JavaScript (Angular, Node.js), Qt-Designer, Shiny, Flask, Docker, Git, REST APIs, CI/CD pipelines, TensorFlow.js, High-Performance Computing (HPC), Embedded Systems, Agile, Scrum
Data Science & Machine Learning:	Statistical analysis, data visualization (Matplotlib, Seaborn, Power BI), machine learning (scikit-learn, XGBoost), deep learning (TensorFlow, PyTorch, Keras), computer vision (OpenCV, scikit-image), generative models, reinforcement learning, dimensionality reduction, unsupervised & supervised learning, MySQL, R, Pandas, NumPy, SciPy
Biomedical Engineering & Medical Imaging:	MRI, CT, PET, SPECT, Optical/Nuclear Medicine, Microscopy, Image reconstruction algorithms, signal processing, Biomedical Instrumentation, Biomechanics, Tissue Engineering (cell culture, scaffold design), design/calibration/maintenance, regulatory compliance, quality assurance
Tools & DevOps:	Label Studio, OpenSeadragon, GitHub, Docker, HPC, testing & deployment pipelines, JIRA, Notion, Confluence
Project Management & Technical Support:	SDLC, IREB (Requirements Engineering), team leadership, stakeholder communication, project coordination, roadmap planning, resource allocation, multidisciplinary collaboration, documentation
Language Skills	English (C2), German (B1), Tamil (Native), Hindi (Fluent), Telugu (Fluent)

Positions of Responsibility

Core Member (TEDx NIT Raipur)

- Lead the team of 6 Executives over a period of 90 days to raise sponsorship of .2 million INR.
- Worked hand in hand with core members of other verticals for smooth conduction of TEDx NIT Raipur.

Convenor (RobotiX Club, NIT Raipur)

- Spearheaded a team of 50+ coordinators and executives for the successful execution of RoboQuiz 2.0 on an online platform.
- Collaborated with campus ambassadors from various colleges, attracting 100+ participants nationwide.